

Ultrafiltration

Final Lab Report

Unit Operations 2 Lab 1

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Group 3



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1. Executive Summary



Ultrafiltration the key operation when it comes to concentrating solutions containing very small sized particles. The objective of this experiment is to better undying understanding of ultrafiltration systems, and to determine the permeability of the hollow fiber membrane with a water and water-milk solution, calculating the filtering effectiveness and the concentration factor. This is done by determining the flux of both liquids at several different outlet pressures and percent recoveries. Data collection is done throughout the experiment determining flux values at varying pressure drops and at 3 different permeate recoveries. In the data analysis section of this report, the relationship of permeate water flux as a function of differential inlet and outlet pressure was modelled for all 3 of the permeate recoveries. It was concluded that there is a linear increase of permeate flux when the pressure drop decreases (higher outlet pressure). Additionally, it was also shown that the overall flux would decrease when operating at higher recovery percentages.



This implies that the optimum pressure settings for the greatest permeate flux is when operating at the lowest percent recovery of permeate, and at the lowest inlet to outlet pressure difference. Effects of concentration polarization during the experiment was nonexistent. A potential reason for this was due to not being able to operate at a high enough permeate percentage. However, additional testing would need to be taken place to insure this theory.



2. Introduction



Ultrafiltration emerged in the 1970s, and it is still a relatively new unit operation in chemical engineering. Ultrafiltration can be used to: concentrate the solute by removing solvent from solution, purify the solvent by removing solute from solution, or separate Solute A from

Solute B by using a membrane that is permeable to only solute A (or vice versa). The ultrafiltration membrane is usually produced from a thin porous plastic film. The pores in the membrane could be anywhere from 10-30 nanometers in diameter. This is just large enough to let the solvent molecules pass through, but not large enough to allow every molecule in the solution to pass through. The use of ultrafiltration is presented in the purification of natural products, the recovery of pharmaceuticals, and has applications for environmental management.

In this experiment, Ultrafiltration was used to concentrate the solute (milk) by removing the solvent (water) from the solution. The ultrafilter was a hollow fiber membrane. The filter was a bundle of small hollow tubes. The water-milk solution became filtered as it flowed through the tubes because the solvent (water) radiated outwards and passed through the pores of the tube [1].

The whole ultrafiltration system was examined, as was its filtering effectiveness. The flux was measured at different pressure drops and for different milk concentrations.

3. Theory

Ultrafiltration can first be explained with a mass balance over the system,

$$\text{Mass original} = \text{Mass concentrate} + \text{Mass permeate}$$

Equation 1: Overall Mass Balance for Ultrafilter

This leads into the concentration factor of the original solution, which is a function of the permeate recovery.

$$CF = \frac{V_{\text{original}}}{V_{\text{concentrate}}} = \frac{V_{\text{original}}}{V_{\text{original}} - V_{\text{permeate}}}$$

$$CF = \frac{1}{1 - R}$$

$$R = \frac{V_{permeate}}{V_{original}}$$

Equation Set 2: Concentration factor of solution

Here, R is the recovery of permeate. The basic equation for permeate flux through the system is

$$v = Q_v \Delta P$$

Equation 3: Flux as a function of pressure

Here, Q_v is the permeability of the membrane. This becomes untrue once a solute is present however, as it will concentrate at the surface of the membrane as shown in the figure below:

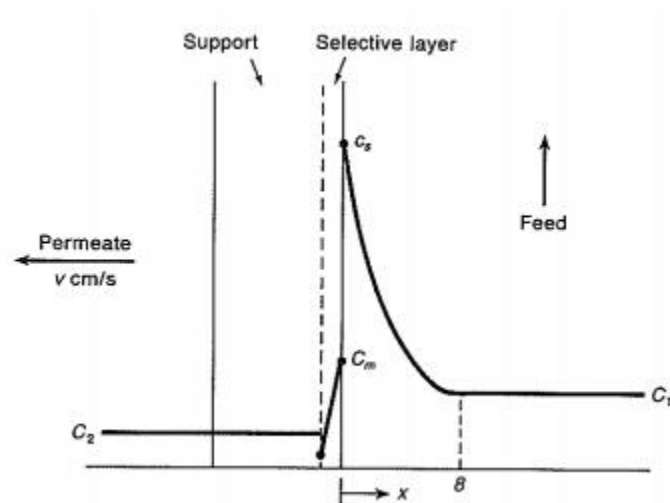


Figure 1: Concentration Gradient at Boundary Layer on Membrane Surface [2]

This higher concentration at the membrane surface, which can be 20-50 times greater than the bulk concentration [3], results in a resistance to flow through the layer that readjusts

itself to allow the convective transport to the membrane surface to be equal to the back-diffusive transport, shown below:

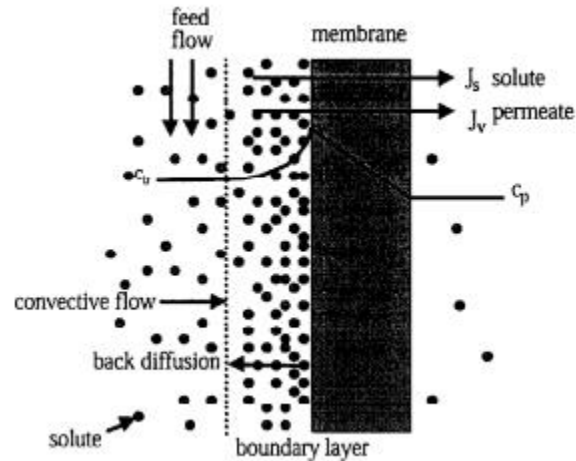


Figure 2: In-depth View of Concentration Polarization [4]

The balance of these two sets of parameters at steady state is

$$C_1 v = -D_v \frac{dC_1}{dx}$$

Equation 4: Steady State Concentration Polarization Model [5]

Solving the differential equation, this leads to the equation for flux in the polarization regime,

$$v = k_c \ln \left(\frac{C_s}{C_1} \right)$$

Equation 5: Dependence of Flux in Concentration Polarization Model [5]

There have been many studies done to prove this correlation, and one such plot is shown below.

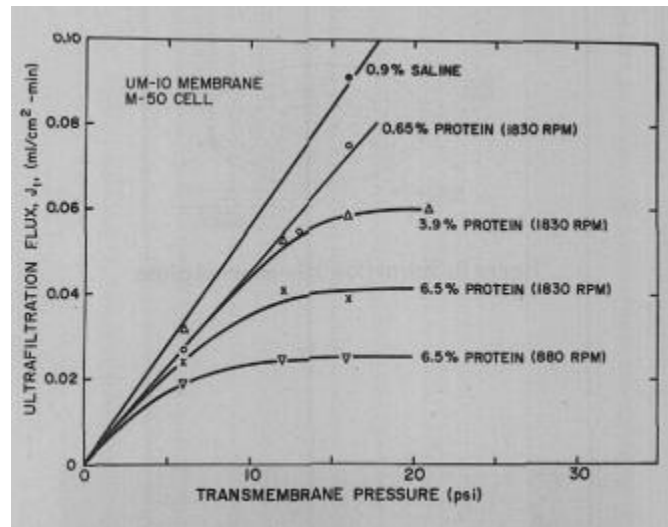


Figure 3: Flux vs Transmembrane Pressure for Bovine Serum Solution [5]

The concentration polarization clearly has a greater influence with higher solute concentration.

Another such test was done for skim milk, similar to what was performed for this lab. This gives an expected trend for a skim milk solution, however this is more concentrated than what is used in lab.

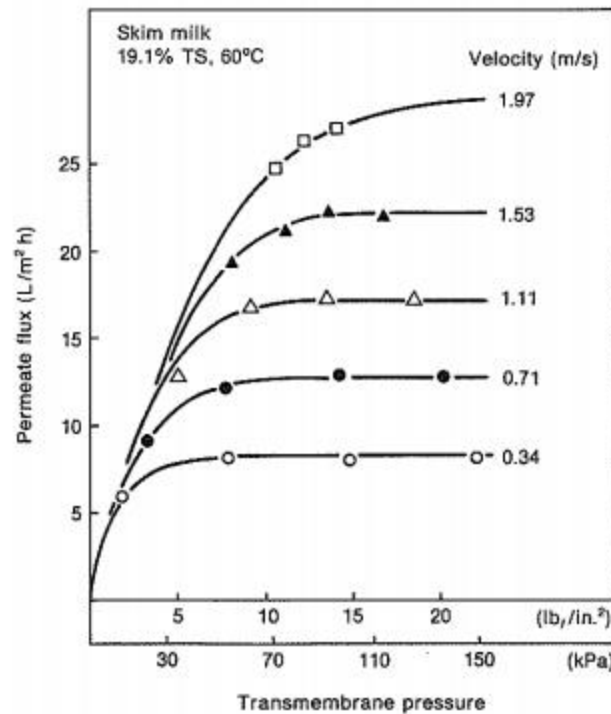


Figure 4: Flux vs Transmembrane Pressure for Skim Milk [6]

4. Results



For the results of this lab, the first thing to address is the basic dependence of the flux through the ultrafiltration on the pressure gradient with no solute present. This is demonstrated by the water flux test.

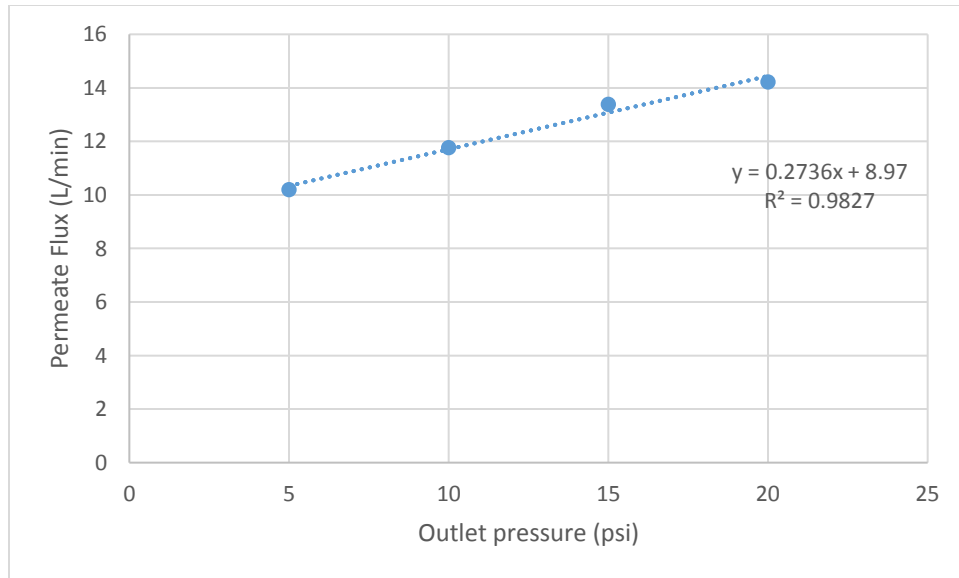


Figure 3: Water Permeate Dependence on Outlet Pressure

The water flux is shown to increase linearly with outlet pressure, leading to the conclusion that the lowest pressure difference, with 25 psi for V1 and 20 psi for V2, is the best. With the whole milk solute present this may change however, based on the effects of concentration polarization previously mentioned in the theory section.

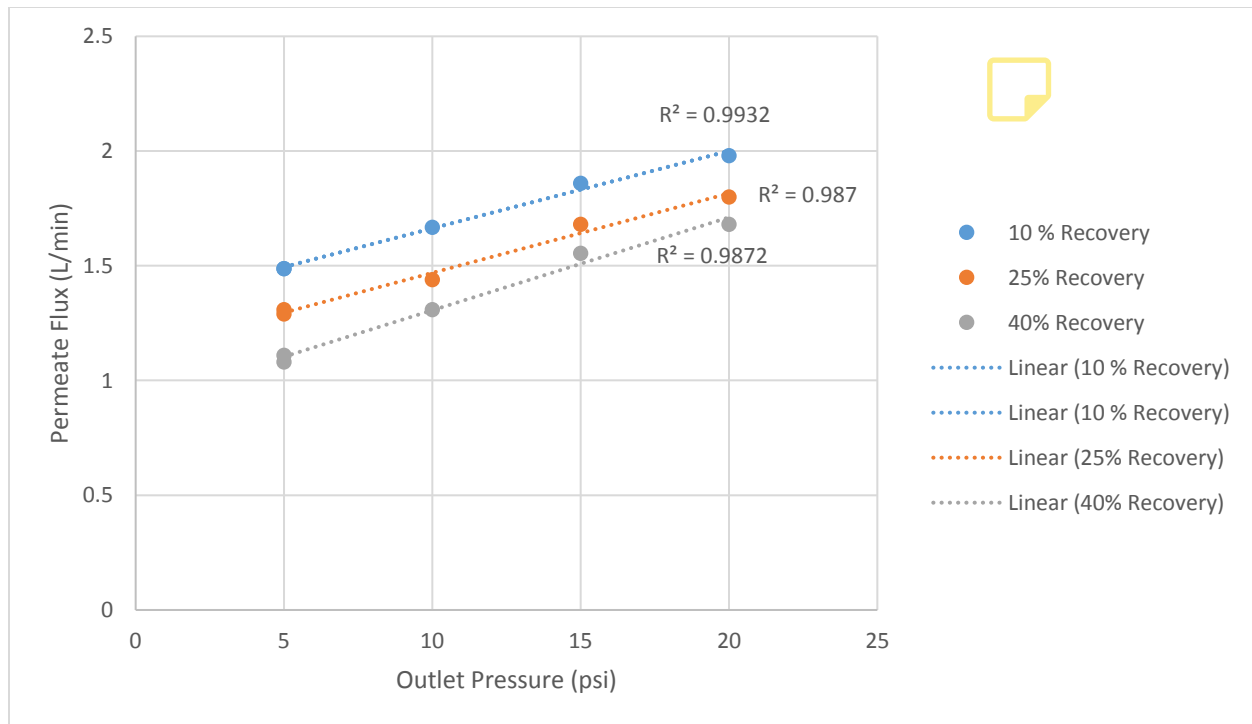


Figure 4: Whole Milk Test Permeate Dependence on Outlet Pressure

This plot shows that the permeate flux is still dependent on outlet pressure at all recoveries studied, contrary to what one would expect considering concentration polarization. Possible reasons for this are explored in the discussion section below.

5. Discussion

As seen in figure 1, the water flux test was used as a baseline to demonstrate the relationship between flux and pressure. There is a strong linear relationship between the two with a correlation value of .9827. This was used to compare to the whole milk test which was done at 3 different recoveries. The recoveries used were 10, 25, and 40%. These recoveries led to concentration factors of 1.11, 1.33, and 1.67 respectively. At each percent recovery, there were 4 different pressures used providing enough data to do a regression. The correlation value at 10, 25, and 40% recoveries were .9932, .987, and .9872. These values indicate a strong correlation

between the flux and the pressure. A possible reason for the lack of concentration polarization was the low concentration of milk in the solution overall. The example plot given in Section 3 was with 19.1% skim milk, compared to this lab with 1 gallon of milk which is approximately 4 liters in 80 liters of water, which gives a percentage of 5.56%, 6.67%, and 8.33% milk in solution at 10%, 25%, and 40% recovery respectively.

6. References

- 1) Woodard, John. "What Is Ultrafiltration? How an Ultrafiltration Membrane Works." *Fresh Water Systems | Ceramic Water Filtration Systems*, Fresh Water Systems, 30 Jan. 2019, www.freshwatersystems.com/blogs/blog/how-an-ultrafiltration-membrane-works.
- 2) W. L. McCabe, J.C. Smith, P. Harriott, "Unit Operations of Chemical Engineering", pp.1037 – 1058, 7th ed. McGraw-Hill, 2005.
- 3) Xiafu Shi, Galit Tal, Nicholas P. Hankins, Vitaly Gitis, Fouling and cleaning of ultrafiltration membranes: A review, *Journal of Water Process Engineering*, Volume 1, 2014, Pages 121-138, ISSN 2214-7144, <https://doi.org/10.1016/j.jwpe.2014.04.003>.
- 4) SS Sablani, MFA Goosen, R Al-Belushi, M Wilf, Concentration polarization in ultrafiltration and reverse osmosis: a critical review, *Desalination*, Volume 141, Issue 3, 2001, Pages 269-289, ISSN 0011-9164, [https://doi.org/10.1016/S0011-9164\(01\)85005-0](https://doi.org/10.1016/S0011-9164(01)85005-0).
- 5) Porter, Mark, "Concentration Polarization with Membrane Ultrafiltration", *Ind. Eng. Chem. Prod. Res. Dev.* Vol. 11, No. 3, pp. 234-248, 1972.

7. Appendix I: Data Tabulation/Graphs

**Water Flux Test**

Table 1: Water Flux Test

Outlet pressure (psi)	Permeate Collected over 10s interval, L	Flux (L/min)
5	1.7	10.2
10	1.96	11.76
15	2.23	13.38
20	2.37	14.22

Whole Milk Test

Table 2: Whole Milk Test

Recovery %

10

Outlet Pressure (psi)	Collected 10s (mL)	Permeate Flux (L/min)
5	248	1.488
10	278	1.668
15	310	1.86
20	330	1.98
5	248	1.488

Recovery %

25

Outlet Pressure (psi)	Collected 10s (mL)	Permeate Flux (L/min)
5	218	1.308
10	240	1.44
15	280	1.68
20	300	1.8
5	215	1.29

Recovery %

40

Outlet Pressure (psi)	Collected 10s (mL)	Permeate Flux (L/min)
5	180	1.08
10	218	1.308
15	259	1.554
20	280	1.68
5	215	1.29

8. Appendix II: Error Analysis

Errors in the results could have occurred for various reasons. The system not being properly cleaned could indirectly affect the flux. For the filters would gradually build up residue and provide blockage to flow, decreasing the flux. While testing the flux, the recommend time was 30 seconds, however, the flow was too great, resulting in measurements of 10 seconds instead. The pump would undergo a pressure drop while measuring the flux, affecting these results. Below room temperature milk could cause fouling of the membrane.

9. Appendix III: Sample Calculations

The sample calculations are taken from outlet pressure 5 psi (water flux test), and recovery 10% (whole milk test).

At an outlet pressure of 5 psi, permeate collected over 10 seconds was 1.7 liters.

$$\text{Flux} = 1.7 * 6 = 10.2 \text{ Liters per minute}$$

At an outlet pressure of 5 psi, with a recovery of 10 %, permeate collected over 10 seconds was 248 milliliters.

$$\text{Flux} = \frac{(248 * 6)}{1000} = 1.488 \text{ Liters per minute}$$

Concentration factor

$$\frac{\text{Original}}{(\text{Original} - \text{Permeate})} = \text{Concentration Factor}$$

% Recovery

$$\frac{\text{Permeate}}{\text{Original}} * 100 = \% \text{ Recovery}$$

10. Appendix IV: Individual Team Contribution

NAME: Matthew Ryan

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	
Pre-Lab Editing	.5	
Experimental Objective (Prelab)	.5	
Apparatus (prelab)	.5	
Materials & Supplies (prelab)	.75	
Procedure (prelab)	.5	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	.1	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	.5	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	6.25	

NAME: Matthew McKinney

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	.5	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	0	
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	

Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	.25	
Total	8.75	

NAME: Matthew Wagener

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	.5	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	.25	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	1	
Discussion (final lab)	0	
Conclusion (final lab)	.5	
References (final and/or prelab)	.25	
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	8.25	

NAME: Joseph Kostenski

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	.5	
Procedure (prelab)	0	

Project Safety Evaluation (prelab)	0	
Final Lab Editing	.25	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	.5	
Error Analysis (final lab)	.5	
Sample Calculations (final Lab)	.5	
Power Point Presentation	0	
Total	7.25	

NAME: Michael Topalis

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	.25	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	1	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	6.25	

